

Tunahan Erbay

Senior Fullstack Software Engineer

Hi, I'm Tuni. I build software that is designed to last. With 6 years of experience across frontend and backend development, I enjoy solving complex technical challenges, improving existing systems, and creating reliable solutions that deliver real value.

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EXPERIENCE

○ Senior Software Engineer

ELCA Informatik AG — Bern, CH

Aug 2022 - Present

Joined ELCA as a Fullstack Software Engineer and was promoted to Senior Software Engineer in 2025. Over time, I took on additional responsibilities as Scrum Master and Lead Developer, contributing to software architecture, technical decision-making, and the successful delivery of enterprise software solutions. Currently, I am working on a new project, helping establish its frontend, backend, and DevOps foundations while driving development across the full technology stack.

- › Promoted from Software Engineer to Senior Software Engineer while taking on additional responsibilities as Scrum Master and Lead Developer.
- › Led agile delivery processes as Scrum Master during a major release, facilitating collaboration between Swiss and offshore development teams.
- › Helped establish a new project from the ground up, including frontend, backend, DevOps, and development processes.

Spring Boot Angular SAP CC WSO2 Git Maven Jenkins Bitbucket Jira Confluence

○ Web Developer

Relaxor — Olten, CH

Jan 2021 - Jul 2021

Built and operated an e-commerce business to gain hands-on experience in web development, digital marketing, and customer acquisition.

- › Built a WooCommerce-based e-commerce platform from scratch, including frontend implementation and visual design.
- › Acquired initial customers through targeted Google Ads and Meta Ads campaigns.

WooCommerce Wordpress JavaScript HTML CSS E-Commerce Marketing Google Ads Meta Ads

○ Founder / Web Developer

Webmo — Olten, CH

Jan 2020 - Present

Developed and maintained WordPress websites and hosting solutions for private individuals and small to medium-sized businesses. Responsible for the technical implementation, customization, maintenance, and long-term support of client solutions.

- › Designed and operated a Linux- and Nginx-based hosting platform for client projects.
- › Developed, customized, and maintained WordPress websites and plugins for more than 20 clients.

WordPress JavaScript HTML CSS Client Support Linux Nginx

EDUCATION

○ Bachelor of Computer Science

University of Basel — Basel, CH

Aug 2018 - Jul 2022

Bachelor's degree in Computer Science with a focus on software engineering, programming, databases, algorithms, and practical software development projects.

○ Swiss Federal Matura

Kantonsschule Olten — Olten, CH

Aug 2014 - Jul 2018

Secondary education with a specialization in Physics and Applications of Mathematics.

SKILLS

BACKEND

Java Spring Boot Python SQL

FRONTEND

Angular TypeScript JavaScript
HTML CSS React

DEVOPS & CLOUD

Git Docker Kubernetes k3s ArgoCD
GitHub Actions Traefik CI/CD
GitOps Maven Jenkins Bitbucket
RHOS

TOOLS

IntelliJ IDEA Visual Studio Code
Jira Confluence Figma Miro

METHODS

Scrum SAFe BPMN
Software Architecture

SOFT SKILLS

Problem Solving Communication
Teamwork Technical Leadership
Analytical Thinking

PROJECTS

tuni.dev — Portfolio & DevOps Platform May 2026
Personal Project — Remote

Built and deployed a personal portfolio website with a full DevOps pipeline, covering containerization, GitOps-based continuous delivery, and cloud infrastructure on Kubernetes.

- › Built a fully responsive portfolio site in React + TypeScript with Tailwind CSS, with dynamic content driven entirely by a single JSON file.
- › Built a CI/CD pipeline with GitHub Actions that builds a Docker image, pushes to GHCR, and auto-updates the Kubernetes manifest on every push.
- › Deployed k3s on a VPS with ArgoCD for GitOps-based delivery — repo changes sync to the cluster automatically.
- › Configured Traefik + cert-manager for automatic SSL across multiple subdomains via Let's Encrypt.
- › Secured the server with a firewall and set up uptime monitoring with a public status page.

Docker Kubernetes k3s ArgoCD GitHub Actions Traefik cert-manager GitOps CI/CD React

Bachelor Thesis: Secure Scuttlebutt Kotlin Library Jan 2022
University of Basel — Basel, CH

Developed a Kotlin-based implementation of the Secure Scuttlebutt (SSB) group chat protocol and transformed it into a reusable library, enabling secure and decentralized group communication within the Tremola ecosystem.

- › Implemented the protocol's core cryptographic functionality, including key derivation, HMAC-based key generation, and secure message encryption/decryption to ensure confidential group communication.
- › Designed and implemented the Slot/Unslot mechanisms and group identifier mapping, enabling secure recipient access control while preserving group privacy in public references.

Kotlin Security Encryption HMAC

Data Integration and Analysis Pipeline Jan 2021
University of Basel — Basel, CH

Designed and implemented a data integration pipeline for collecting, transforming, and analyzing heterogeneous datasets from multiple sources.

- › Analyzed 4 GB of U.S. crime and economic data drawn from multiple sources and mixed datatypes, using SQL and Pandas to integrate them into a single coherent dataset. Performed data cleaning, integration, and visualization.

SQL Python Data Engineering ETL Data Visualization Data Analysis

Linux Command-Line Integration Tool Oct 2020
University of Basel — Basel, CH

Reimplemented core Linux shell functionality and file system operations in C, including support for remote file system integration.

- › Developed a minimal Gnome-Terminal-style shell in C, reimplementing 20+ core commands and adding support for mounting remote servers over SSH so they appear as regular directories.

C Systems Programming Filesystems SSH

Decentralized Messaging System May 2020
University of Basel — Basel, CH

Developed a secure decentralized messaging application enabling peer-to-peer communication without relying on centralized servers.

- › Built an end-to-end encrypted chat application UI in Python using Tkinter, implementing decentralized peer-to-peer communication over BACnet without relying on a central server.

Python Encryption Networking Peer-to-Peer Security

Modular LED Gaming Console Jan 2020
University of Basel — Basel, CH

Designed and built a modular LED-based gaming console from scratch, including both hardware and software components.

- › Designed and built a custom multiplexed LED matrix with simulated circuitry and hand-soldered components, then programmed a plug-in game framework with joystick input in Arduino/C++.

C C++ Circuit Design Embedded Systems Hardware Design Arduino

○ JabRef User Interface Extension

[University of Basel](#) — Basel, CH

Developed a user interface extension for the open-source project JabRef, enhancing customization and accessibility features.

- › Developed and implemented a user interface extension for JabRef, a widely used open-source Java application for managing BibTeX and BibLaTeX databases. The extension provided additional customization options, including themes, color schemes, and templates, while addressing accessibility requirements and enhancing usability.

Java

Scrum

Accessibility

○ Multiplayer Gorillas Remake

May 2019

[University of Basel](#) — Basel, CH

Designed and developed a multiplayer remake of the classic Gorillas game, focusing on game physics, networking, and real-time synchronization.

- › Implemented deterministic physics, collision handling, and a real-time peer-to-peer networking layer in Java for synchronized multiplayer.

Java

Game Physics

Networking

CERTIFICATIONS & COURSES

- Developing Generative AI Applications on AWS

Fast Lane Jan 2026

Generative AIAWS
- Communication

ELCA Feb 2025

Communication
- Professional Scrum Master I

Scrum.org Aug 2024

AgileScrumcertified
- Kubernetes Native Development with Quarkus

Red Hat May 2024

KubernetesQuarkusArchitecture
- Usability/UX Introduction

ELCA Sep 2023

UXUsability
- Quality in Software Design

ELCA May 2023

Software DesignQuality
- BPMN Method & Style

Stephan Fischli Mar 2023

BPMNcertified
- Angular - The Complete Guide

Udemy Sep 2022

AngularTypeScriptJavaScript
- Leading SAFe - Certified SAFe 6 Agilist

Scaled Agile, Inc. Mar 2025

SAFeAgilecertified
- Spring & Hibernate for Beginners

Udemy Nov 2024

SpringHibernateSpring Boot
- Problem Solving

ELCA Jun 2024

Problem Solving
- The Role of Scrum Master

ELCA Oct 2023

Scrum
- Angular Advanced Training

Jonas Bandi Jun 2023

AngularTypeScript
- Code Quality

ELCA May 2023

Code Quality
- Application Security Fundamentals

ELCA Feb 2023

Application Security